MILITARY SPECIFICATION

# HALOPROGIN SOLUTION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 This specification covers Haloprogin Solution.

1.2 Classification. The solution covered by this specification shall be the following sizes (see 6.2):

Size 1 - 10 ml
Size 2 - 30 ml

## 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

| | |
|---|---|
| PPP-C-186 | Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals (see 6.3). |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | Marking for Shipment and Storage. |

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)



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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN PHARMACEUTICAL ASSOCIATION PUBLICATION

National Formulary.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

U.S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and
Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

Pharmacopeia of the United States.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be a non-greasy, non-staining solution of Haloprogin, 10 mg per ml, in a homogeneous vehicle of diethyl sebacate and alcohol.

3.2 Assay of the finished solution.

3.2.1 Haloprogin. Shall contain not less than 0.90 percent and not more than 1.10 percent (W/V) when determined as specified in 4.5.1.

3.2.2 Alcohol. Shall contain not less than 71.1 percent and not more than 78.7 percent alcohol (V/V) when determined as specified in 4.5.2.

3.3 Identification of Haloprogin in the finished solution. Ultraviolet spectrum of the sample shall show maxima at the same wavelengths and conforms to that of a Haloprogin Reference Standard (See 4.5.1).



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3.4 Specific gravity of the finished solution shall be not less than 0.838 and not more than 0.852 at $25^{o}/25^{o}C$, when determined as specified in 4.5.3.

3.5 Ingredients.

3.5.1 Haloprogin.

3.5.1.1 Melting range - Shall melt between $110^{o}C$ - $114^{o}C$ when determined as specified in 4.6.1.1.

3.5.1.2 Loss on drying - Shall not exceed 0.5 percent when dried in a vacuum oven for 4 hrs. at $60^{o}C$.

3.5.1.3 Free acid - Solution shall remain yellow when prepared as specified in 4.6.1.2.

3.5.1.4 Heavy metals - Shall not exceed 50 parts per million when determined as specified in 4.6.1.3.

3.5.1.5 Residue on ignition - Shall not exceed 0.1 percent when determined using the U.S.P. method except that the temperature of the muffle oven shall be $600^{o}C \pm 25^{o}C$.

3.5.1.6 Infrared identification. Shall comply with 4.6.1.4.1.

3.5.1.7 Ultraviolet identification. Shall comply with 4.6.1.4.2.

3.5.1.8 Assay - Shall assay to contain not less than 95.0 percent and not more than 102.0 percent $C_9H_4Cl_3IO$ when determined as specified in 4.6.1.5.

3.5.2 Diethyl sebacate.

3.5.2.1 Refractive Index - Shall be not less than 1.4350 and not more than 1.4370 when determined at $20^{o}C$ using a suitable refractometer with an accuracy of $\pm$ 0.0001 units.

3.5.2.2 Acid value - Shall not exceed 1 when determined as specified in 4.7.1.

3.5.2.3 Infrared Identification - Shall comply with 4.7.2.

3.5.2.4 Assay - Shall contain not less than 98 percent $C_{14}H_{26}O_4$ when determined as specified in 4.7.3.



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3.5.3 Other ingredients. All other ingredients entering into the manufacture of the solution shall be of U.S.P. or N.F. quality or, if not included in either of these compendia, the ingredients shall be of the highest pharmaceutical quality and suitable for use in the solution.

3.6 Workmanship. Material and its containers shall be free from defects which detract from their appearance or may impair their serviceability.

3.7 Expiration dating period. The expiration dating period shall be 36 months.

3.7.1 Delivery schedule. Not less than 33 months of the expiration dating period shall remain at the time of delivery to the Government.

3.8 If (i) the quality assurance representative submits samples to the laboratory at the Defense Personnel Support Center, or (ii) the bidder or offeror is required by the terms of the Schedule or otherwise to submit samples to said laboratory, a quantity of 5 grams of Haloprogin used in the manufacture of each lot of Haloprogin solution and 0.5 gram of Haloprogin Reference Standard, accompanied by a certificate of analysis listing all test results, shall be submitted with the finished product to the Technical and Quality Assurance Division, Directorate of Medical Materiel, Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, PA 19101, Attention: Quality Assurance Branch.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period (potency period) specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the items to the Government.



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4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Lot. For purpose of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of finished solution produced from one formulation, subjected to the same compounding and manufacturing operations, and filled into final containers.

4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

Table I. Sampling

| For visual examination | Inspection level | AQL (percent defective) | Unit of product |
|---|---|---|---|
| Major | S-2 | 1.0 | Immediate container with controlled dropper tip |
| Minor | II | 2.5 | |

4.3.1 For raw material and end item testing. Non-compliance with any of the following is cause for rejection.

Table II. Raw material and end item testing

| Characteristic | Requirement paragraph | Inspection level* | Minimum sample |
|---|---|---|---|
| Assay (Haloprogin) | 3.2.1 | S-2 | 5 |
| Assay (Alcohol) | 3.2.2 | S-2 | 5 |
| Identification | 3.3 | S-2 | 5 |
| Specific gravity | 3.4 | S-2 | 5 |
| Ingredients: | | ** | |
| Haloprogin | 3.5.1 | - | - |
| Diethyl sebacate | 3.5.2 | - | - |

* Where possible, the sample shall be used for two or more tests.

** Sufficient quantities of each of the raw materials shall be selected from each lot to perform tests indicated in applicable test methods. Failure of any sample to comply with test requirements shall be cause for rejection of the lot represented.

4.4 Examination. The finished solution shall be examined to determine compliance with all requirements of this specification. Nonconformance will be permitted to the extent indicated in 4.3 and 4.3.1.



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4.4.1 Classification of defects. Examination will be conducted in accordance with the following classification of defects:

Categories and defects*

| | |
|---|---|
| Major | |
| 101 | Solution not free from foreign matter. |
| 102 | Solution has crystal formation. |
| 103 | Solution not homogeneous. |
| 104 | Solution not non-greasy or non-staining. |
| 105 | Controlled dropper tip defective or missing. |
| Minor | |
| 201 | Exterior of immediate container is not free from contents or foreign matter. |
| 202 | Label not free from defacing marks and tears. |

* Inspection is not restricted to the classified possible defects listed above.

4.5 Assay of the finished solution.

4.5.1 Haloprogin.

Standard solution. Accurately weigh approximately 100 mg of Haloprogin Reference Standard into a 100-ml volumetric flask. Dissolve in and dilute to volume with methanol. Pipet 5.0 ml into a 50-ml volumetric flask and dilute to volume with methanol.

Procedure

Pipet 1.0 ml of sample solution into a 100-ml volumetric flask and dilute to volume with methanol. Concomitantly determine the absorbances of sample and standard solutions with a suitable recording spectrophotometer in a 1-cm cell from 400 to 250 nm, using methanol as the blank.

Calculation:

$$\frac{Au}{As} \times \left(\frac{\text{mg standard}}{100} \times \frac{5}{50}\right) \times \left(\frac{100}{1}\right) = \text{mg/ml}$$

$$\text{mg/ml} \times \frac{100}{1000} = \text{percent required amount haloprogin per ml}$$

Where: $Au = \frac{\text{absorbance sample (298 mu - absorbance 320 mu)}}{}$

$As = \text{absorbance standard (298 mu - absorbance 320 mu)}$



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4.5.2 Alcohol. Shall be determined by the U.S.P. method or as an alternate by a suitable gas chromatographic method.

4.5.3 Specific gravity. Shall be determined by the U.S.P. pycnometer method, or as an alternate using either a Christian Becker Style SG-2, or Westphal Balance, with a sensitivity of 1 mg or less.

4.6 Ingredients.

4.6.1 Haloprogin powder.

4.6.1.1 Melting range - Using the U.S.P. apparatus, introduce the sample into the bath at 80°C and permit the temperature to rise 3°C per minute to 108°C, at which point reduce the rate of increase to 1° - 2°C per minute.

4.6.1.2 Free acid - To 10 ml of acetone add 2 drops of methyl red solution (0.1 percent in alcohol). If required add 0.01N sodium hydroxide solution dropwise until the solution just turns yellow. To the above solution add 1.00 gram of haloprogin powder and dissolve. Add 3.0 ml of 0.01N sodium hydroxide and observe whether solution changes in color.

4.6.1.3 Heavy metals - Weigh (within 10 mg) 0.5 grams of sample into a beaker, add 30 ml acetone and 2 ml 1N acetic acid. Agitate to dissolve the sample (solution B).

Pipet 2.5 ml of lead standard solution prepared in accordance with the U.S.P. procedure for heavy metals, containing 0.01 mg Pb/ml into a second beaker, add 2 ml 1N acetic acid and 30 ml acetone, and mix (solution A).

Adjust the pH of each of the above solutions to 3.0 - 4.0 using 1N acetic acid or ammonia T.S. as required. Transfer solutions with the aid of purified water to separate matched 50 - ml Nessler tube, add 1 drop of sodium sulfide solution T.S., dilute to volume with purified water, mix, and compare darkness of color after 5 minutes.

(Note - If sample solution is not clear filter immediately through #42 Whatman paper prior to the waiting period.)

4.6.1.4 Identification.

4.6.1.4.1 Infrared spectrum. The infrared absorption spectrum of a potassium bromide dispersion of it, previously dried, exhibits maxima only at the same wavelengths as that of a similarly prepared Reference Standard. A copy of the IR spectrum of the Reference Standard shall be submitted by the contractor and shall be kept on file in the Technical and Quality Assurance Division, Directorate of Medical Materiel, Defense Personnel Support Center for use if samples are submitted.



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4.6.1.4.2 Ultraviolet spectrum - The ultraviolet spectrum of a solution of a final concentration of approximately 0.1 mg per ml in alcohol exhibits maxima at the same wavelengths and conforms to that of a similarly prepared solution of Haloprogin Reference Standard. A copy of the UV spectrum of the Reference Standard shall be submitted by the contractor and shall be kept on file in the Technical and Quality Assurance Division, Directorate of Medical Materiel, Defense Personnel Support Center for use if samples are submitted.

4.6.1.5 Assay - Accurately weigh approximately 250 mg of haloprogin into a flask and dissolve in 15 ml of methanol. Slowly add a solution of 200 mg of sodium borohydride in 10 ml of methanol to the flask with shaking and while cooling under water. Continue until the foam disappears (approximately 2 minutes). Permit to stand 10 - 15 minutes and add 30 ml of purified water (a white precipitate forms). Add 40 ml concentrated hydrochloric acid (The precipitate remains). Cool, add 7 ml chloroform and shake. The aqueous solution beocmes clear and the chloroform layer becomes slightly pink in color. Titrate with M/30 $KIO_3$ volumetric solution until the chloroform layer becomes colorless.

(Note - shortly after titrating both layers assume a brown color. In the aqueous phase the brown turns to yellow while the chloroform color remains brown-pink. At the endpoint the chloroform color disappears rapidly.)

Calculation:

$$\frac{\text{ml } KIO_3 \times F \times 24.094 \times 100}{\text{sample weight (mg)}} = \text{Percent haloprogin}$$

Where F = Normality Factor

4.7 Diethyl sebacate.

4.7.1 Acid value. Accurately weigh (within 0.2 mg) approximately 56.4 grams of sample into a 250 ml Erlenmeyer flask. Dissolve the sample in 50 ml of alcohol-ether solution prepared and previously neutralized in accordance with the USP procedure. Reflux gently for ten minutes. Cool to room temperature, add 1 ml of Phenolphthalein T.S., and titrate the sample solution with 0.1N Sodium Hydroxide Volumetric Solution until a pink color persists after shaking for thirty seconds.

Calculation:

$$\frac{\text{ml NaOH} \times \text{Norm.} \times 56.1}{\text{Sample weight (grams)}} = \text{acid value}$$



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4.7.2 Infrared identification. The infrared absorption spectrum of a thin, uniform film in sodium chloride plates with air as a reference exhibits maxima only at the same wavelengths as that of a similarly prepared Reference Standard. A copy of the IR spectrum of the Reference Standard shall be submitted by the contractor and shall be kept on file in the Technical and Quality Assurance Division, Directorate of Medical Materiel, of the Defense Personnel Support Center for use if samples are submitted.

4.7.3 Assay (saponification value).

Ester Reagent.

Weigh (within 0.1 gram) 10.5 grams KOH, dissolve in 9 ml of purified water, and cool to room temperature. Separately mix 180 ml of diethylene glycol and 14.4 ml of alcohol. Add with continuous stirring the solution of KOH to the diethylene glycol-alcohol mixture.

Procedure

Accurately weigh (within 0.2 mg) approximately 1.2 grams of diethyl sebacate into an Erlenmeyer flask. Use a second flask without sample as the blank. Treat each flask as follows: Add 25.0 ml of ester reagent and mix. Attach a condenser and reflux on a steam bath for 5 minutes. Cool to room temperature and rinse the condensers with purified water into the flasks. Add 2 drops of phenolphthalein T.S. and titrate with 0.5N HCl Volumetric Solution to the absence of a red endpoint. Calculate the saponification value.

4.8 Preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table III. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 (defects per hundred units).

Table III. Examination of preparation for delivery

| Examination | Defect |
|---|---|
| Containers | Not as specified. |
| Quantity | Quantity in interior packages and shipping containers not as specified. |
| Packaging and Packing Materials | Any component missing or damaged. |
| Workmanship | Inadequate application of components such as incomplete closure of container flaps, tape improperly applied, or containers distorted. |
| Markings | Omitted, incorrect, illegible, improper size, location, sequence, or method of application. |

NOTE: The above examination is only applicable to the unit, intermediate and exterior containers.



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5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with all applicable paragraphs of PPP-C-186 and as specified herein.

5.2 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.2.1 Unit of issue. One bottle containing 10 ml of size 1 material or 30 ml of size 2 material, as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. The immediate container shall be a commercially available bottle with a controlled dropper tip, considered acceptable to the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit package. At the option of the contractor, a unit package may be furnished. If a unit package is furnished, each unit shall be packaged as specified in 5.2.5 of PPP-C-186.

5.2.2.3 Intermediate package. Twenty-four units of size 1 or size 2 shall be packaged as specified in 5.2.7.1 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.3.1 Level A. Twenty-four intermediate packages (576 units) of size 1 or twelve intermediate packages (288 units) of size 2 material shall be packed in an exterior container as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Twenty-four intermediate packages (576 units) of size 1 or twelve intermediate packages (288 units) of size 2 material shall be packaged in an exterior container as specified in 5.4.3.1.1 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.



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5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.

(b) the expiration date.

5.4.1.2 Unit package. Each unit package label shall bear the following:

(a) label information in accordance with commercial practice.

(b) the National Stock Number.

NOTE: (1) The National Stock Number shall not be preceded by any prefix.

(2) Strip labeling may be used.

(3) Label adhesion test shall not apply.

(4) If a unit package is not supplied, the National Stock Number shall appear on the immediate container label.

5.4.2 Intermediate package. Each intermediate package shall be marked in accordance with MIL-STD-129. Lot (control) number, contract or purchase order number and name of contractor shall be shown. Type I Shelf Life Marking as specified in MIL-STD-129 shall also be shown. Marking shall include the legend:

"STORE AT CONTROLLED ROOM TEMPERATURE ($59^{o}$ - $86^{o}$ F)".

5.4.3.3 Exterior container. Exterior containers shall be marked in accordance with MIL-STD-129. The lot (control) number shall be shown. Type I Shelf Life Marking as specified in MIL-STD-129 shall also be shown. Marking shall include the legend:

"STORE AT CONTROLLED ROOM TEMPERATURE ($59^{o}$ - $86^{o}$ F)".



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6. NOTES

6.1 Ordering data. Procurement documents should specify the following;

(a) Title, number and date of this specification.
(b) the National Stock Number.
(c) Selection of applicable levels of packaging and packing (See 5.2 and 5.3).

6.2 This specification covers the following item(s):

| National Stock No. | Item Identification |
|---|---|
| 6505-00-118-2211 | HALOPROGIN SOLUTION, 1%, 10 ml |
| 6505-00-118-2308 | HALOPROGIN SOLUTION, 1%, 30 ml |

6.3 All references to PPP-C-186 refer to Interim Federal Specification PPP-C-00186B which is mandatory for use until superceded by a revision.

6.4 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1395
cs



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6. PREPARING ACTIVITY

# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-H-37247 HALOPROGIN SOLUTION

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

**DD FORM 1 JAN 72 1426** REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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DEFENSE SUPPLY AGENCY
DoD - 304




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PENALTY FOR PRIVATE USE $300

HEADQUARTERS, DEFENSE PERSONNEL SUPPORT CENTER
ATTN: DIRECTORATE OF MEDICAL MATERIEL, CODE ATT
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